

PROBLEM SET / 04

Evidence-aware organizational graph inference

Conservative relationship inference and navigable organizational knowledge

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Developed after the original submission and grading dates. Status: In progress.

Abstract

Communication records can suggest authority, collaboration and membership, but these are different relationships. This workbench keeps their types and provenance separate while building an inspectable organization graph. The implementation combines staged intake, conservative local inference, immutable SQLite snapshots, reversible analyst review and a React interface with bounded semantic zoom. [1] [2]

A deterministic synthetic corpus with 10,000 people and 76,680 messages was imported and analyzed in a recorded local run. It produced 7,489 selected reporting links and retained 2,511 unresolved people. Read-only traversal reached every person through formal and inferred groupings. These results demonstrate workflow and navigability on authored data, without treating raw scores as probabilities or generated labels as independent accuracy evidence. [3] [4]

Problem and design objective

The SCADS problem concerns deriving organizational knowledge from partial evidence and supporting exploration as the graph grows. The implemented objective is to preserve the distinction between what a source states, what the baseline infers and what an analyst reviews. A visually complete tree must not conceal unresolved managers or convert communication frequency into direct supervision. [1] [2] [6]

Implemented contribution

The working system integrates evidence spans, typed authority relations, sparse candidate generation, revision-aware review and scalable navigation. Intake profiles the proposed corpus before saving it. The graph can be explored through departments, communication groups, reporting branches and individual connections without drawing every person at once. [1] [2]

Evidence class: fictional communications and source hierarchies, engineering checks and recorded local timings. Ranking scores are explicitly uncalibrated; synthetic labels are excluded from inference inputs.

1. Research framing and relation types

Prabhakaran and Rambow study power relations between participants in written dialog. That task motivates language-based authority signals, but relative power within an interaction is not identical to a direct reporting edge. This prototype explicitly separates contextual approval authority, direct reporting, formal membership and communication communities. [6] [8]

Guo et al. distinguish classification scores from calibrated probabilities of correctness and evaluate post-processing calibration. Their work is a methodological reference, not evidence that temperature scaling applies automatically to this heuristic graph baseline. Calibration must be fitted and evaluated on labels representative of the relation, selection rule and target population. [9]

Formal target

Let P be people, M observed messages and A_s attributed source assertions. The inference target is a set of candidate directed relations R subset of $P \times \text{Types} \times P$ with evidence and temporal scope. A selected primary reports_to edge is a separate projection, not the complete set of candidates. The system may leave a person unresolved. [2]

Candidate: (subject, relation, object, score, evidence, time)

Origin: source | model | analyst

Review: unreviewed | accepted | rejected

Selection != truth; raw score != calibrated probability

The main research question is whether communications support useful direct-report reconstruction under partial labels and domain transfer while preserving uncertainty. A second question concerns exploration: can all relevant people and claims remain reachable without requiring an unreadable global layout? These questions need separate evidence; browser navigation cannot validate manager accuracy.

Proposed inference estimand: correctness of selected direct-manager edges at a declared time among people with independently verified labels, alongside selection coverage and abstention. Relative authority and team membership require their own labels.

2. Method and implemented architecture

Layer	Implemented responsibility
Intake	Parse canonical JSON, message/roster CSV and mail formats; profile quality and preview a bounded source graph.
Inference	Generate sparse candidates from communications and explicit self-report language; retain unresolved people.
Evidence	Keep message spans, communication aggregates, source assertions and analyst references distinct.
Persistence	Store immutable snapshots, durable jobs and append-only review events in SQLite.
Exploration	Serve snapshot-bound pages and bounded graphs through FastAPI to React/TypeScript.

The canonical assertion includes subject, relation, object, origin, evidence IDs, review status and temporal bounds. Primary and matrix reporting can be represented separately. Raw scores and calibration status are distinct fields, and a reviewed replacement does not acquire a model probability merely because it appears in the same chart. [2]

Isolate labels from inference

The inference service strips evaluation labels, existing assertions and source evidence before sending entities and messages to its baseline engine. The fixture labels can describe authored structure without becoming hidden inputs. This is necessary for a meaningful software check, although shared organization and language templates still limit any accuracy interpretation. [1] [2] [4]

Review is a versioned action

Accept, reject, replace and undo operations require the current workspace revision and preserve their history. Rejecting a relation continues to matter across refreshed inference. Exported canonical packages can preserve historical projections and evidence; package restoration requires an empty workspace. Snapshot comparison evaluates semantic relationships instead of merely comparing assertion IDs. [2] [3]

Intake inspection is staged in memory before an explicit build. The preview reports accepted records, missing-field coverage and a bounded source graph. It does not portray imported proposals as established authority. Exact inventory counts and visible processing stages let a user understand what will be saved. [2] [3]

3. Conservative baseline in detail

The implemented baseline caps behavior-only scores below the default selection threshold. Let f and r be eligible directional interaction counts, $s=f/\max(1,\text{total outgoing})$, and $\rho=\min(f,r)/\max(1,f,r)$. Its communication score is the following deterministic heuristic; the coefficients are implementation choices, not learned or calibrated parameters. [10]

```
b = 0 when there are no eligible pair records; otherwise:
b = min(0.52, 0.08 + 0.20*s + 0.14*rho
+ 0.10*min(1, log(1+f+r)/log(11)))
```

An explicit sender self-report gives a higher candidate score based on distinct claim text and interaction count, capped at 0.96. A contradictory negated report caps the score at 0.30. Quoted text is conservatively excluded from sender claims. Approval requests produce a separate `higher_authority_than` relation rather than a `direct-manager` score. [10]

```
rank candidates per person by score, then stable ID
select a primary candidate only if:
no explicit contradiction
highest score >= 0.55
highest score - second score >= 0.08
dated-view requirements are satisfied
otherwise retain the person with an unresolved reason
```

A useful consequence follows directly: pure communication behavior cannot exceed 0.52, so under the default 0.55 threshold it cannot independently select a manager. This intentionally favors abstention over converting frequent contact into formal authority. The reported number of selected links therefore reflects the designed evidence profile of the synthetic corpus, not an empirically calibrated confidence level. [10]

Sparse structure and display cost

Candidate generation retains each person's top eight observed communication contacts plus all explicit positive/negative reporting targets, instead of evaluating every person pair. If C is the retained candidate set, scoring work is tied to messages and C rather than an unavoidable dense $|P|^2$ matrix. The browser separately bounds its local scene. Actual throughput still depends on communication density, parsing and storage. [1] [2] [10]

4. Worked example

The synthetic entry path defaults to 10,000 people. Its reviewed preview measured 76,680 messages, 45 units, two shared mailboxes and 60,657 directed communication links. Evaluation labels remained separate. The workflow draws inspect -> profile -> plan -> database -> infer -> ready; explanatory replay is labeled and does not perform another inference run. [3]

Interaction	Recorded behavior	Analytical purpose
Enter a department	Move from aggregate scope to reporting branches	Keep organization-wide navigation readable.
Focus a person	Display selected manager, alternatives and evidence	Connect the chart to its claims.
Continue zooming	Expand from 29 to 79 to 199 links	Reveal context without shifting existing card positions.
Inspect a reporting marker	Show source/model origin and review/calibration state	Avoid reading visual certainty as statistical confidence.
Return to a prior scope	Restore camera and expanded context	Preserve the analyst's place in the investigation.

Bound the display, preserve reachability

The interface retains a rolling metadata window of at most 240 cards and renders at most 200 nearby cards. A bounded response cache supports return navigation. Reporting branches are presentation groups derived from selected links, not new formal teams. Explicit unassigned buckets keep unresolved people reachable. [1] [3]

Read-only traversal of the live synthetic snapshot reached all 10,000 people through both chart lenses: 476 formal scopes over 501 pages, and 1,077 inferred scopes over 1,082 pages. Edge closure was checked so every displayed link referred to displayed people. These are coverage and consistency observations rather than browser frame-rate measurements. [3]

Figure 1. The interaction sequence reflects recorded browser checks. Formal units, inferred communication groups and reporting branches remain differently labeled throughout the journey. [3]

5. Recorded results and interpretation

10,000-person communication run	Observed value
Messages / entities	76,680 / 10,047
Generation / import	0.711 s / 9.199 s
Inference and publication	46.979 s
Selected / unresolved people	7,489 / 2,511
Formal and inferred groups	107

These values describe one local Windows run, with source and corpus fingerprints preserved. The synthetic generator uses deterministic templates and authored labels; the outcome is not an accuracy, calibration or unlabeled-transfer result. Unresolved people are an explicit output rather than nodes forced into a complete tree. [4]

Separate source-only benchmark	10,000 mixed	100,000 mixed
Import	2.749 s	77.945 s
List page warm p95	27.311 ms	120.228 ms
Substring search warm p95	45.378 ms	754.730 ms
Focused graph warm p95	40.723 ms	42.893 ms

The second table measures Python/SQLite reads on source-only hierarchies, using seven warm repetitions and a 200-node graph budget. Its nearest-rank p95 is descriptive for that small sample; caches were not flushed. HTTP, browser rendering, dense communication processing and inference quality are outside this benchmark. [5]

Verification and interpretation

The recorded current intake build passed 144 Python tests, 99 frontend checks and the production build. Browser verification covered source preview, saved-workspace reuse, isolated import/build, cancellation and responsive layout. The evidence supports a durable, inspectable workflow and bounded navigation; it does not establish that communications reveal real managerial authority with calibrated confidence. [3] [7]

6. Proposed inference and user study

PROPOSED STUDY. The synthetic implementation evidence is not a substitute for independently labeled organizations or transfer validation. The following is the planned research design.

Question	Design	Evidence to report
Direct-report reconstruction	Partial-label source organization; held-out people/time blocks	Candidate recall, selected precision, coverage and abstention
Authority versus reporting	Separate labels for contextual power and direct management	Relation-specific errors and confusion
Calibration	Fit on validation labels; assess untouched matched scope	Reliability curves, Brier score and calibration error
Transfer	Freeze model before different organization; acquire audit labels	Domain shift, errors and applicability limits
Exploration utility	Counterbalanced analyst tasks with bounded versus baseline views	Correct answers, active time, navigation and frame-time traces

A transfer corpus with no labels can test operability, but cannot establish accuracy or calibration by itself. Obtain an independent audit sample before making those claims. Keep departments or temporal blocks together where leakage through repeated correspondents would inflate performance. A communication edge is not an independent observation if many edges arise from the same person or thread. [\[11\]](#)

Compare at least a communication-only baseline, explicit linguistic baseline and a learned candidate when data justify it. Select thresholds on validation data, then report risk versus coverage on the untouched test set. Calibration should target the selected edge distribution, not merely arbitrary candidate pairs; abstention changes that distribution.

For the interface, preserve information equivalence across conditions and randomize task order. Measure correctness of evidence-based answers before time savings. Record viewport, browser, graph density, network and hardware; reachability and closure tests remain correctness checks, while frame time and analyst effort are separate empirical outcomes.

7. Roadmap and limits of inference

Phase	Output	Decision rule
Qualify relations and data	Authorized corpora with explicit direct-report/time labels	Separate authority, membership and influence targets.
Train and calibrate	Frozen feature/model selection and matched calibration split	Compare quality at stated selection coverage.
Transfer and review	Independent target audit and reversible identity/membership review	Retain uncertainty where evidence is insufficient.
Evaluate exploration	Density-aware browser tests and analyst study	Claim navigability, performance and utility separately.

This extends an existing local workbench. Identity reconciliation and richer review can change graph topology, so later evaluations must bind both model and identity-decision versions. Exported histories should keep original evidence and analyst interventions distinguishable, enabling a study of whether corrections improve subsequent inference rather than merely altering a displayed chart. [2] [7]

Threats to validity

The present generator shares authored organization structure and language templates with its labels. Label exclusion prevents direct input leakage, but it does not create an independent distribution. Explicit self-reports are unusually clean compared with real archives. Selection totals therefore demonstrate conservative behavior under the fixture design, not performance on a workforce. [4]

Storage benchmarks use source-only hierarchies and seven warm reads. Their p95 equals a small-sample order statistic and should not be interpreted as a service-level guarantee. Conversely, a graph that renders rapidly may still communicate unjustified certainty. Unresolved nodes, origin labels and evidence access are part of the scientific meaning of the interface. [3] [5]

Conclusion

The prototype connects conservative typed inference with a durable, navigable evidence workbench. Its implemented abstention policy and bounded display provide a concrete basis for testing reconstruction and analyst utility. The scientific claim remains feasibility and inspectability on authored data, with real inference quality reserved for the proposed labeled study.

8. Reproducibility and research record

Use a separate database for each reproduced corpus. The default small fixture offers a quick review/export workflow; the deterministic generator also supports the documented 10,000-person case. Save source fingerprints and record environment and load when comparing timings. [1] [4]

```
python -m orggraph demo --people 10000 --output runs/demo/corpus.json --generate-only
python -m orggraph --db runs/demo.sqlite3 demo --people 10000
python -m pytest -q
npm --prefix frontend test
```

Run from the nested project directory after the documented dependency setup. The benchmark script and raw result file specify workload shapes, read methods and cache assumptions. The verification report links the browser evidence and distinguishes successive software milestones; test totals should always be attributed to the corresponding snapshot. [1] [3] [5]

No raw score is presented as a probability. Software tests, graph reachability, storage timing and real-world inference quality remain separate quantities in this publication.

A reproducible research record

The public nested project source contains implementation, documented contracts and retained development evidence. The publication evidence manifest records SHA-256 fingerprints for each local source cited in this paper. External references link to primary publications or official technical documentation, verified for this edition.

A future rerun should record the exact commit, dependency/runtime identities, input hashes, configuration and environment, plus each attempt and its outcome. Keep generated outputs separate from source and mark any changed implementation as a new condition. Reproduction of a software check does not automatically reproduce a model judgment, benchmark distribution or independent human label.

Authorship and project status

Boomer Rawlings developed this work individually after the original submission and grading dates. The current project is In progress. The implemented methods and recorded observations in this paper are separated from the explicitly proposed research protocol and roadmap. No group-project, graded-submission or completed-research claim is made.

References

Repository references identify the evidence used for the implementation and result claims. Primary literature supports the research framing; its published results are not attributed to this project.

[1] [Implemented workbench and synthetic corpus](#)
IMPLEMENTATION.md

[2] [Canonical data and API contract](#)
docs/implementation-contract.md

[3] [Recorded software and browser verification](#)
docs/verification.md

[4] [10,000-person communication-corpus result](#)
docs/synthetic-10000-results.json

[5] [Synthetic storage benchmark](#)
docs/benchmark-results.json

[6] [Inference research and evaluation distinctions](#)
research/inference.md

[7] [Completion evidence and acceptance scope](#)
docs/acceptance.md

[8] [Prabhakaran and Rambow \(2014\). Predicting Power Relations between Participants in Written Dialog from a Single Thread. ACL, pp. 339-344.](#)
<https://aclanthology.org/P14-2056/>

[9] [Guo et al. \(2017\). On Calibration of Modern Neural Networks. ICML; arXiv:1706.04599.](#)
<https://arxiv.org/abs/1706.04599>

[10] [Implemented conservative inference baseline](#)
orggraph/inference.py

[11] [Corpus qualification and evaluation research](#)
research/data-and-evaluation.md